

Communication Node

Connectivity & Gateways | Product brochure

Communication Node



The Communication Node enables traditional analog and wired sensors to be monitored remotely over LoRaWAN networks. By digitizing analog signals and transmitting them wirelessly, it turns legacy instruments such as strain gauges, crack meters, piezometers, load cells, and temperature probes into connected monitoring assets without forcing a full sensor replacement.

Key benefits

- Third-party sensors, one integrated system
- Wireless LoRaWAN connectivity for remote and large-scale sites
- Multi-year battery autonomy with configurable acquisition settings
- Full remote configuration and alarms via the monitoring platform
- IP67-rated standalone design for harsh outdoor environments
- Multi-input analog interface

Supports up to 8 analog sensor interfaces

- 4-20 mA
- mV / Wheatstone bridge
- Voltage inputs (5 V, 12 V)
- Potentiometer
- NTC thermistor
- Pt100 / Pt1000
- Vibrating wire sensors

Engineering highlights

Legacy sensor integration: Connects third-party analog sensors without replacing existing instrumentation.

LoRaWAN transport: Reliable long-range data transmission in remote sites.

Autonomous operation: Battery-powered deployment with intelligent acquisition logic.

What this sensor measures

- Probe data from connected analog sensors, converted into engineering units such as mm, deg C, or kPa.
- Vibrating wire frequency, converted to strain, pressure, displacement, or equivalent units.
- Internal or external temperature used for data correction and correlation.

Technical specifications

Specification	Value
Communication	LoRaWAN / LTE / Wi-Fi
Input	8 analog channels
Ingress protection	IP67

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